ANDERSON JUNIOR COLLEGE
2009 PRELIMINARY EXAMINATION
HIGHER 2

NAME:____ **PDG:** _________

CHEMISTRY **9746/02**

Paper 2 Structured Questions **18 September 2009**

1 hour 30 minutes

Candidates answer on the Question Paper.

Additional Materials : Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and PDG on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

A Data Booklet is provided.

The number of marks is given in brackets [ ] at the end of each questions or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use		
Section A	Q1	
	Q2	
	Q3	
	Q4	
	Q5	
	Total	60

This paper consists of **13** printed pages including this cover page.

1 Aluminium and sulfur are elements in the third period of the Periodic Table. There are many useful applications of these elements and their compounds.

(a) Explain how the first ionisation energy of aluminium will differ from that of sulfur.

....

....

.... **[2]**

(b) (i) Describe the difference in the structure and bonding between the oxides of aluminium and sulfur.

....

....

....

(ii) Describe the reaction, if any, of the oxides of aluminium and sulfur with water. Give balanced equations for any reactions that occur and suggest the pH values of any resulting solutions.

....

....

....

....

....

.... **[5]**

(c) In catalytic hydrogenation of alkenes, a catalyst known as Raney nickel is commonly used. Raney nickel is produced when a block of nickel-aluminium alloy is treated with concentrated sodium hydroxide. This treatment, called "activation", dissolves most of the aluminium out of the alloy. The aluminium-containing product of this reaction is the same as that obtained from the reaction between aluminium oxide and sodium hydroxide. The porous structure left behind has a large surface area, which gives high catalytic activity.

When a 1.00 g sample of this alloy was reacted with concentrated sodium hydroxide, 0.71 dm^3 of hydrogen gas was produced at 40 ^oC and a pressure of 1 atm.

(i) Construct a balanced equation for the reaction between aluminium and sodium hydroxide.

....

[1]

(ii) Calculate the number of moles of hydrogen gas produced.

Number of moles of hydrogen gas = ……………………………. mol **[2]**

(iii) Hence calculate the percentage of aluminium in the sample of alloy.

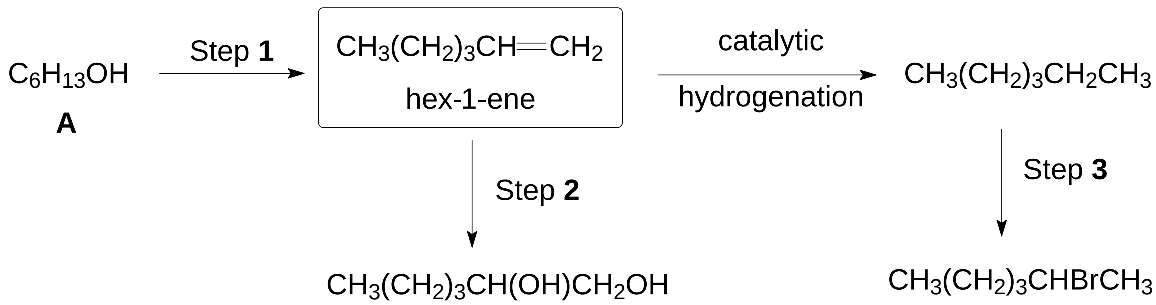
Percentage of aluminium in sample = ……………….. % **[2]**

(d) Catalytic partial hydrogenation of vegetable oils to produce margarine produces *trans fat* which increases a person's risk of coronary heart disease. *Trans fat* is a lipid molecule that contains one or more double bond in a *trans* configuration. One example is elaidic acid, $CH_3(CH_2)_7CH=CH(CH_2)_7COOH$.

Draw the structure of the *trans isomer* of elaidic acid.

[1]

(e) Hex-1-ene is a co-monomer used in the production of polyethene, which is widely used to manufacture plastic shopping bags. Some reactions of hex-1-ene are shown below.



(i) Name and describe the type of reaction mechanism that occurs in step **3**.

Type of mechanism :

(ii) State the reagents and conditions for:

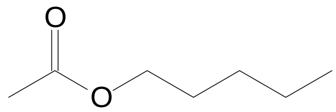
Step **1** :

Step **2** :

(iii) Compound **A** can exist as a pair of stereoisomers. Draw the structure of compound **A**.

[6]
[Total: 19]

2 Compound **B** is found in pear oil which is an artificial flavouring. The structure of **B** is shown below.

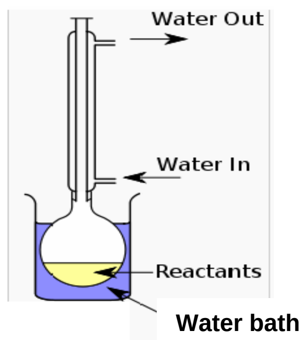


compound **B**

Compound **B** can be synthesised by reacting a carboxylic acid and an alcohol. The equation for the reaction is shown below.

$$\text{RCOOH}\,(l) \quad + \quad \text{R'OH}\,(l) \quad \rightleftharpoons \quad \mathbf{B}\,(l) \quad + \quad H_2O\,(l)$$

In an experiment, 3 mol dm^{-3} each of the carboxylic acid and alcohol are refluxed together using the set-up shown below. At equilibrium, 1.06 mol dm^{-3} of the alcohol was left in the equilibrium mixture.



(a) **(i)** Write an expression for the equilibrium constant, K_c.

(ii) Calculate the value of K_c for this reaction.

K_c = ………………………… **[3]**

(b) The experiment was repeated using the same initial concentrations of the carboxylic acid and alcohol. During the experiment, the water in the condenser stopped flowing and the smell of pear was detected from the top of the condenser.

State and explain, if any, how this problem would affect

(i) the composition of the equilibrium mixture,

....

....

....

(ii) the value of K_c.

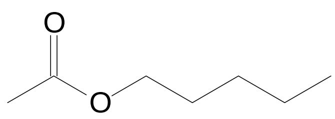
....

....

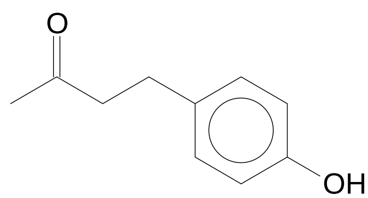
[4]

(c) Compound **C** is used to to give a raspberry flavour while compound **D** is used to impart apple flavours to food. The structures of compounds **C** and **D** are shown below.

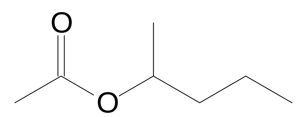
compound **B**



compound **C**



compound **D**



Suggest methods by which the compounds **C** and **D** could be distinguished from compound **B** by chemical tests, giving reagents, conditions and observations for each compound.

The distinguishing test(s) may rely on the preliminary breaking-up of the compounds and the subsequent testing of the reaction products.

(i) Compound **B** and **C**

(ii) Compound **B** and **D**

[5]
[Total: 12]

3 **(a)** Calcium oxide, CaO, is one of the products formed from the thermal decomposition of calcium carbonate, $CaCO_3$, in a special kiln at about 500 ^oC to 600 ^oC.

(i) Write a balanced equation for the thermal decomposition of calcium carbonate.

....

(ii) Using relevant data from the *Data Booklet,* describe and explain how the thermal stability of zinc carbonate differs from that of calcium carbonate.

....

....

....

....

(iii) Draw a diagram to show the shape of the CO_3^{2-} ion.

[5]

(b) Calcium oxide is used in the making of glass and porcelain.

The lattice energy of calcium oxide can be calculated from a Born-Haber cycle using the following information and other relevant data found in the *Data Booklet*.

Enthalpy change of atomisation of calcium	+ 147 kJ mol^{-1}
Enthalpy change of formation of calcium oxide	- 635 kJ mol^{-1}
Sum of first and second electron affinities of oxygen	+ 657 kJ mol^{-1}

(i) Explain, with the aid of an equation, what is meant by the *lattice energy of* calcium oxide.

....

....

....

(ii) Construct a Born-Haber cycle for calcium oxide and use the cycle to calculate the lattice energy of calcium oxide.

∴ Lattice energy of calcium oxide = ………….............. kJ mol^{-1}
[5]

(c) Group II carbonates decompose on heating. In an experiment, an impure sample of $BaCO_3$ was heated till it decomposes. The resulting product was dissolved in dilute nitric acid. The solution was found to contain Ba^{2+}, Fe^{2+} and Pb^{2+} ions.

A series of steps were carried out to separate the three ions. Describe the observations and give the identity of the cation precipitated in steps **1** and **3**.

Step 1 : Addition of excess NaOH (aq) to solution.

Observations : ....

Cation precipitated : ....

Step 2 : Filter and retain filtrate for **step 3**.

Step 3 : Addition of excess dilute HC*l* (aq) to filtrate.

Observations : ....

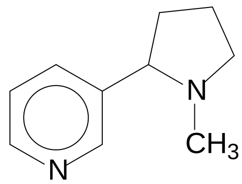
Cation precipitated : ....

[4]
[Total: 14]

4 Nicotine is a naturally occurring liquid alkaloid, which can be extracted from the leaves of the tobacco plant. It is the compound that is responsible for the potent effect of tobacco on the human body. The structure of nicotine is shown below.

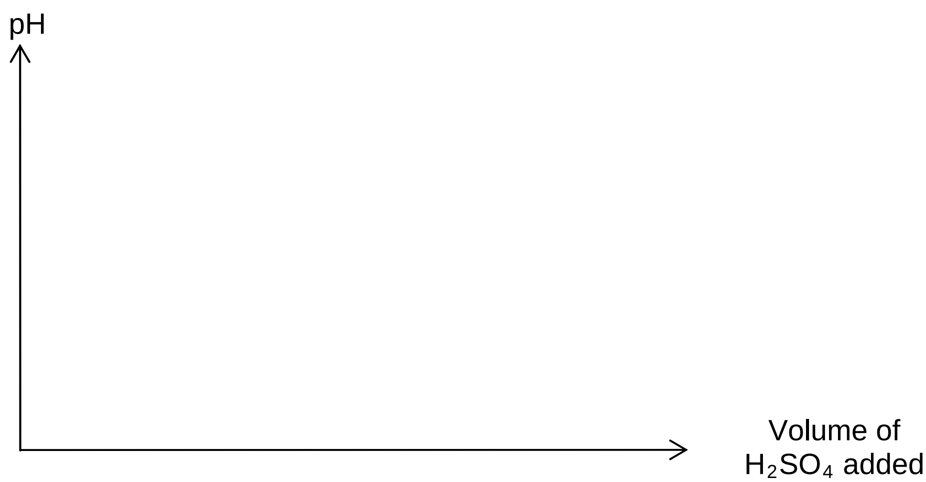
pK_{b1} = 5.98

pK_{b2} = 10.88



15 cm^3 of a 0.10 mol dm^{-3} solution of nicotine was titrated against 0.10 mol dm^{-3} dilute H_2SO_4.

(a) Using the data above, sketch how the pH changes during this titration on the following axes.

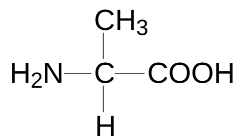
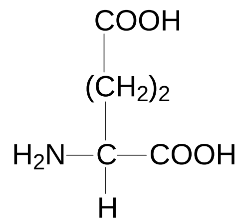
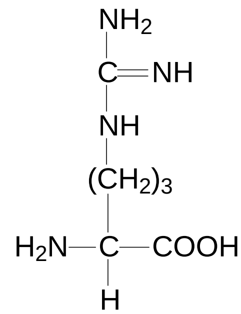


[3]

(b) Draw the structure of the organic compound formed at the **<u>first</u>** equivalence point.

[1]

(c) In an experiment, the inhibitory effect of nicotine on the transport of amino acids in placenta is investigated. The amino acids are alanine, glutamic acid and arginine.

Amino acid			
	alanine	glutamic acid	arginine
Isoelectric point	6.10	3.22	10.74

Before the experiment can be conducted, separation of a mixture of these three amino acids using electrophoresis was carried out. The mixture was placed in the centre of a gel plate buffered at pH = 6.10. The two ends of the gel plate were then connected to a power source until separation was achieved.

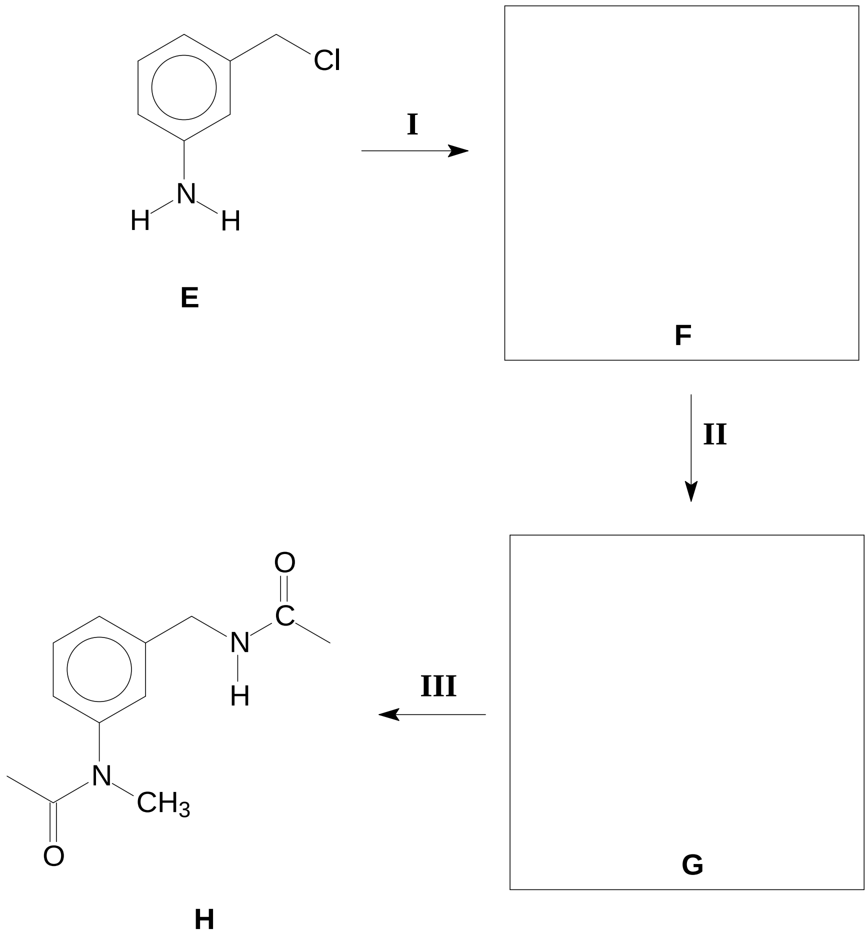
Identify the amino acid at positions **1**, **2** and **3** on the gel plate at the end of the separation.

	1	**2**	**3**	
negative electrode	______________	______________	______________	positive electrode

[3]

[Total: 7]

5 Compound **H** can be synthesised from compound **E** via a series of steps.



(a) State the reagents and conditions for :

Step **I** : ....

Step **II** : ....

Step **III** : ....

[3]

(b) Draw the structures of the intermediates, **F** and **G**, in the boxes above. **[2]**

(c) Write a balanced equation for each of the following reactions, showing clearly the structural formulae of the organic product(s).

(i) Compound **E** with aqueous bromine.

(ii) Compound **H** with hot aqueous potassium hydroxide.

[3]

[Total: 8]